

If there are no external electromagnetic forces (or if the external electromagnetic forces are time independent), the classical model conserves energy, so that it does not have any hyperbolic orbits. On the other hand, in time dependent (e.g. periodic) external fields, many interesting models (e.g. ion traps, mirror magnet machines) have many hyperbolic orbits [Gho96, Kaj22].

In this paper, we show that near hyperbolic orbits of the unperturbed model (59), i.e. $\varepsilon = 0$ in (60)², and if the perturbed model avoids the singularities and it has bounded delays, Theorem 4.8 applies. Moreover, constructed solutions are similar to those hyperbolic orbits for the unperturbed model (59), as long as $0 < \varepsilon \ll 1$ in (60). See Theorem 7.3 for a precise formulation.

Before proceeding to the detailed analysis, let us make some comments on the equations and their physical properties.

1. The expressions defining the forces are algebraic expressions (arithmetic operations and square roots). They have singularities when there are collisions ($q_i(t) = q_j(t)$ for some $i \neq j$) or when a particle reaches the speed of light ($|\dot{q}_i(t)| = c$ for some i).

We will assume that the unperturbed solutions we consider are a finite distance away from these singularities so that the expressions given the second derivatives and given the positions and velocities are smooth functions in a neighborhood of the unperturbed solution. See Definition 7.2.

2. The delays τ_{ij} or advances σ_{ij} as in (60) involve solving implicit equations that involve the pairwise trajectories $q_i(t)$ and $q_j(t)$.

The physical meaning of the delays/advances are the time it takes a light signal to travel from the particle i to the particle j . Since the speed of the particles is bounded away from the speed of light, this time exists and is unique. Nevertheless, finding the delay requires to solve an equation that depends on the pairwise trajectories. See (60).

Fortunately, for the method used in this paper, the main property needed is that τ_{ij} and σ_{ij} are uniformly smooth assuming that the trajectories are smooth (and that they are away from the singularities).

Note that in general $\tau_{ij} \neq \tau_{ji}$ even if $\tau_{ij} - \tau_{ji} = O(\varepsilon^2)$. Indeed, after expanding the solution of (60) up to first order, i.e.

$$\tau_{ij}(t) = \varepsilon |q_i(t) - q_j(t)| + \varepsilon^2 (q_i(t) - q_j(t)) \cdot \dot{q}_j(t) + O(\varepsilon^3). \quad (62)$$

(hint: It is easy to consider the expansions for τ_{ij}^2 obtained by squaring both sides of (60) and express the square of length as an inner product). The above derivation (62) of an approximate form of the delay is purely formal. Note that it is only valid for differentiable q and that the error incurred in the approximation depends on higher derivatives of q . Of course in a delay equation, modifying the form of the delay is a very singular perturbation so, substituting the approximation above may lead to equations with different solutions.

In Theorem 4.8 the main objects are sets of uniformly differentiable q . In these sets of uniformly differentiable functions, the approximations in (62) are uniform (in a norm involving one less derivative than the uniform).

3. Theorem 4.8 also applies to many modifications of the model and produces solutions. For example, ignoring The Figtgerald contractions [Ver16] or changing the delay by its state dependent approximation (62) [Dri84] (they are formally second order in ε). The exact solutions of these models obtained applying Theorem 4.8 will be approximate solutions of (59).

In theoretical Physics, there are also other methods to produce approximate solutions using formal power series [CCdL20, GdLY25, MS78] or using numerical approximations.

²For $\varepsilon = 0$, the delays and advances vanish so that (59) is an ODE.